448 Midterm 1 Project Report

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RQ's 1,2,4 were:

1. Can we identify which LLM made a response?
2. Does the prompt help identify the LLM?
4. What characteristic differentiate LL

Dataset

Used public database from: https://huggingface.co/datasets/lmarena-ai/arena-human-preference-140k/viewer

This was a little of an older dataset with models that aren't quite as new, and some things may have changed about each of Claude's, Gemini's, and Qwen's outputs in recent year, but the methods and results should be similar enough to suffice.

Took about 50k of the original 136k and cleaned up columns and rows to only include the models being compared. Took about 3k per model for training a ~6k was used in training, ~1.4k in validation, ~1.3k in testing.

CNN/RNN Implementation:

Used a CNN and an LSTM in PyTorch to predict whether a response came from Claude, Gemini, or Qwen. Ran each model using just the prompt, just the response, and both together, giving six experiments total.

Converted the text into tokens, keeping up to 128 tokens from the prompt and 256 from the response. Built the vocabulary from training data only and represented each token with 64 learned values. The CNN looked for patterns in groups of 3, 4, and 5 tokens, using 64 filters for each size. The LSTM read the text in order and used a 64-value hidden state to make its prediction. Both used 30% dropout during training.

Trained in batches of 64 using AdamW, a learning rate of 0.001, and cross-entropy loss. Allowed up to 12 epochs, stopping after three without improved validation macro-F1. Kept the best validation checkpoint and measured its accuracy and macro-F1 on the separate test set.

Results:

Tested both architectures using the prompt, response, and both together.

Model	Text used	Accuracy	Macro-F1
CNN	Prompt	33.75%	0.3179
CNN	Response	93.34%	0.9334
CNN	Both	94.00%	0.9401
LSTM	Prompt	32.72%	0.3272
LSTM	Response	84.75%	0.8472
LSTM	Both	84.75%	0.8476

Response-only predictions were well above the roughly 33% expected from guessing. Prompt-only predictions stayed near that level, suggesting the prompt alone gave little useful information. Adding the prompt improved CNN accuracy by only 0.67 percentage points and did not change LSTM accuracy. The uncertainty analysis did not show a reliable benefit from adding prompts.

Additional analysis:

Compared response lengths and formatting across the three families. Median response lengths were about 214 words for Claude, 646 for Gemini, and 434 for Qwen. A simple classifier using length alone reached 52.85% accuracy, while one using ten style features reached 77.35%.

Removed punctuation and newline tokens from the text given to the already-trained models. CNN accuracy dropped to 37.31%, and LSTM accuracy dropped to 39.08%. This suggests sensitivity to formatting, although removing tokens also changes sequence length and position. These results do not prove that formatting alone explains the original accuracy.

Limitations:

Used one specific version from each family and one training seed, so results may differ with newer models or another dataset. Responses were limited to their first 256 tokens, which shortened about 77% of test responses. Related examples were grouped to reduce leakage, but similar paraphrases could still remain. Performance across different task domains was not tested.

Lessons and experience:

I really had to learn the real meat and potatoes of both CNN and RNN, I did use AI, as I understood a lot of the concepts, but the actual putting it into practice was something I struggled with. I would’ve had no idea where to start putting the lessons from class into practice